

# Лабораторная работа №15

## Администрирование локальных сетей

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Ромицына А. Р.

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Российский университет дружбы народов, Москва, Россия

# Информация

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- Ромицына Анастасия Романовна
- НПИбд-02-23 Студ. билет: 1132236132
- Российский университет дружбы народов
- 1132236132@pfur.ru

# Вводная часть

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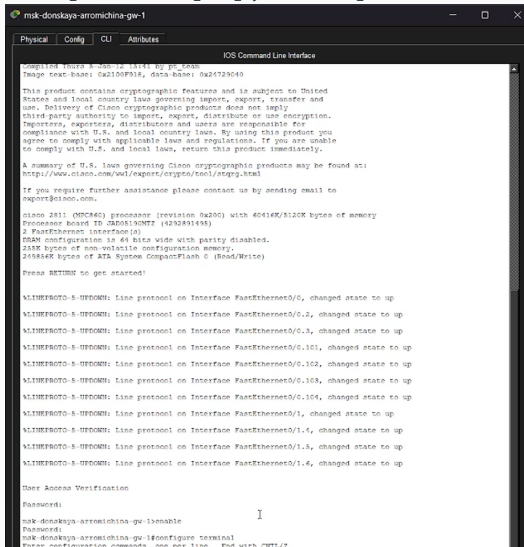
- Настроить динамическую маршрутизацию между территориями организации

## Основная часть

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# Настройка msk-donskaya-gw-1

- Настройка маршрутизатора msk-donskaya-gw-1



```
msk-donskaya-arromichina-gw-1
Physical Config CLI Attributes
IOS Command Line Interface
Copyright 1986-2013 Cisco and/or its affiliates. All rights reserved. Cisco Confidential
Image text-base: 0x2100F018, data-base: 0x24729C40

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
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A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wd1/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

cisco 2811 (MPC8440) processor (revision 0x200) with 40416K/5120K bytes of memory
Processor board ID JAD08190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 44 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
245856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.6, changed state to up

User Access Verification

Password:
```

# Проверка msk-donskaya-gw-1



```
msk-donskaya-aromichina-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

msk-donskaya-aromichina-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 3 secs, Minimum LSA arrival 1 secs
Number of external LSA 0, Checksum Sum 0x000000
Number of opaque AS LSA 0, Checksum Sum 0x000000
Number of Database external and opaque AS LSA 0
Number of Database external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
Area BACKBONE(0)
  Number of interfaces in this area is 8
  Area has no authentication
  SPF algorithm executed 1 times
  Area ranges are
    Number of LSA 1, Checksum Sum 0x00312a
    Number of opaque link LSA 0, Checksum Sum 0x000000
    Number of Database LSA 0
    Number of indication LSA 0
    Number of Database LSA 0
    Flood list length 0

msk-donskaya-aromichina-gw-1#sh ip ospf neighbor

msk-donskaya-aromichina-gw-1#sh ip route
Codes: L - local, C - connected, S - static, B - BGP, M - mobile, R - RDP
D - EIGRP, EX - EIGRP external, O - OSPF, TA - OSPF inter area
NI - OSPF NSSA external type 1, NI - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

0.0.0.0/0 is variably subnetted, 18 subnets, 4 masks
C 10.128.5.0/24 is directly connected, FastEthernet0/0.3
L 10.128.5.1/32 is directly connected, FastEthernet0/0.3
C 10.128.1.0/24 is directly connected, FastEthernet0/0.2
L 10.128.1.1/32 is directly connected, FastEthernet0/0.2
C 10.128.8.0/24 is directly connected, FastEthernet0/0.101
L 10.128.8.1/32 is directly connected, FastEthernet0/0.101
C 10.128.4.0/24 is directly connected, FastEthernet0/0.102
L 10.128.4.1/32 is directly connected, FastEthernet0/0.102
C 10.128.5.0/24 is directly connected, FastEthernet0/0.103
L 10.128.5.1/32 is directly connected, FastEthernet0/0.103
C 10.128.6.0/24 is directly connected, FastEthernet0/0.104
L 10.128.6.1/32 is directly connected, FastEthernet0/0.104
C 10.128.255.0/30 is directly connected, FastEthernet0/1.3
L 10.128.255.1/32 is directly connected, FastEthernet0/1.3
C 10.128.255.0/30 is directly connected, FastEthernet0/1.4
L 10.128.255.1/32 is directly connected, FastEthernet0/1.4
C 10.128.255.0/30 is directly connected, FastEthernet0/1.4
L 10.128.255.1/32 is directly connected, FastEthernet0/1.4
C 10.128.5.0/24 [1/0] via 10.128.254.4
E 198.51.100.0/24 is variably subnetted, 3 subnets, 2 masks
C 198.51.100.0/24 is directly connected, FastEthernet0/1.4
L 198.51.100.1/32 is directly connected, FastEthernet0/1.4
* 0.0.0.0/0 [1/0] via 198.51.100.1

msk-donskaya-aromichina-gw-1#
```

Рис. 1: Проверка состояния протокола OSPF на маршрутизаторе msk-donskaya-gw-1



## Настройка msk-q42-gw-1

- Настройка маршрутизатора msk-q42-gw-1

[illegible]

# Настройка msk-hostel-gw-1



```
msk-hostel-gw-1
Physical Config GUI Attributes
ICS Command Line Interface

65536K bytes of flash-simulated non-volatile configuration memory.
Base Ethernet MAC Address      + C960-0C25-39E6
Hardware serial number         + 73-9473-09
Power supply part number       + 341-0124-00
Hardware serial number         + C011075070
Power supply serial number      + C051154670B
Model revision number          + 20
Hardware revision number        + 20
Model number                   + MS-C3560-24PS-E
System serial number           + C011075070
Tag Assembly Part Number       + 800-34300-04
Tag Assembly Revision Number    + 80
Version 10                     + Y06
CMT Code Number                + COM11008AC
Feature Board Revision Number   + 0001

Switch  Ports  Model  SW Version  SW Image
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* 1 24 MS-C3560-24PS 12.2(37)SE1 C3560-ADVIPSERVICESK

Cisco IOS Software, C3560 Software (C3560-ADVIPSERVICESK9-M), Version 12.2(37)SE1, RELEASE SOFTWARE (x64)
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Thu 08-Jul-07 22:12 by pt_team

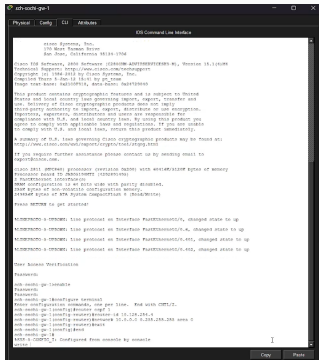
Press RETURN to get started!

VLINE0-0-CPU000: Interface Vlan202, changed state to down
VLINE0-0-CPU000: Interface Vlan302, changed state to down
VLINE0-0-CPU000: Interface Vlan402, changed state to up
VLINE0-0-CPU000: Interface Vlan502, changed state to up
VLINE0-0-CPU000: Interface GigabitEthernet0/1, changed state to up
VLINE0-0-CPU000: Line protocol on Interface GigabitEthernet0/1, changed state to up
VLINE0-0-CPU000: Line protocol on Interface Vlan202, changed state to up
VLINE0-0-CPU000: Line protocol on Interface Vlan302, changed state to up
VLINE0-0-CPU000: Interface FastEthernet0/1, changed state to up
VLINE0-0-CPU000: Line protocol on Interface FastEthernet0/1, changed state to up

Over Access Verification
Password:
msk-hostel-gw-1#enable
msk-hostel-gw-1#configure terminal
msk-hostel-gw-1(config)#router ospf 1
msk-hostel-gw-1(config-router)#router-id 10.121.214.4
msk-hostel-gw-1(config-router)#router-id 10.121.214.5
msk-hostel-gw-1(config-router)#no
```

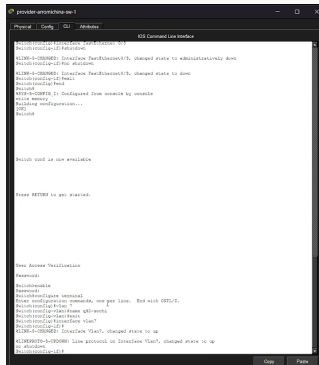
Рис. 2: Настройка маршрутизирующего коммутатора msk-hostel-gw-1

# Настройка sch-sochi-gw-1



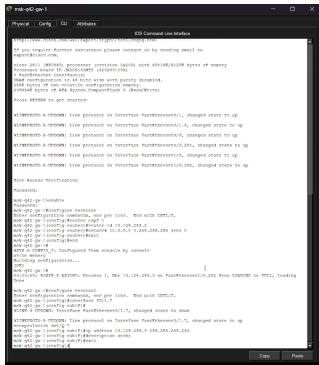
**Рис. 3:** Настройка маршрутизатора sch-sochi-gw-1

## Настройка provider-sw-1



**Рис. 4:** Настройка интерфейсов коммутатора provider-sw-1

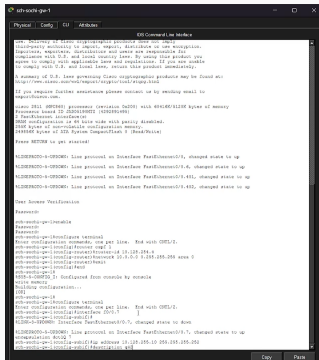
# Настройка msk-q42-gw-1



**Рис. 5:** Настройка маршрутизатора msk-q42-gw-1



# Настройка sch-sochi-gw-1



**Рис. 7:** Настройка маршрутизатора sch-sochi-gw-1



**Рис. 8:** Проверка



## Вывод

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- Мы смогли настроить динамическую маршрутизацию между территориями организации.